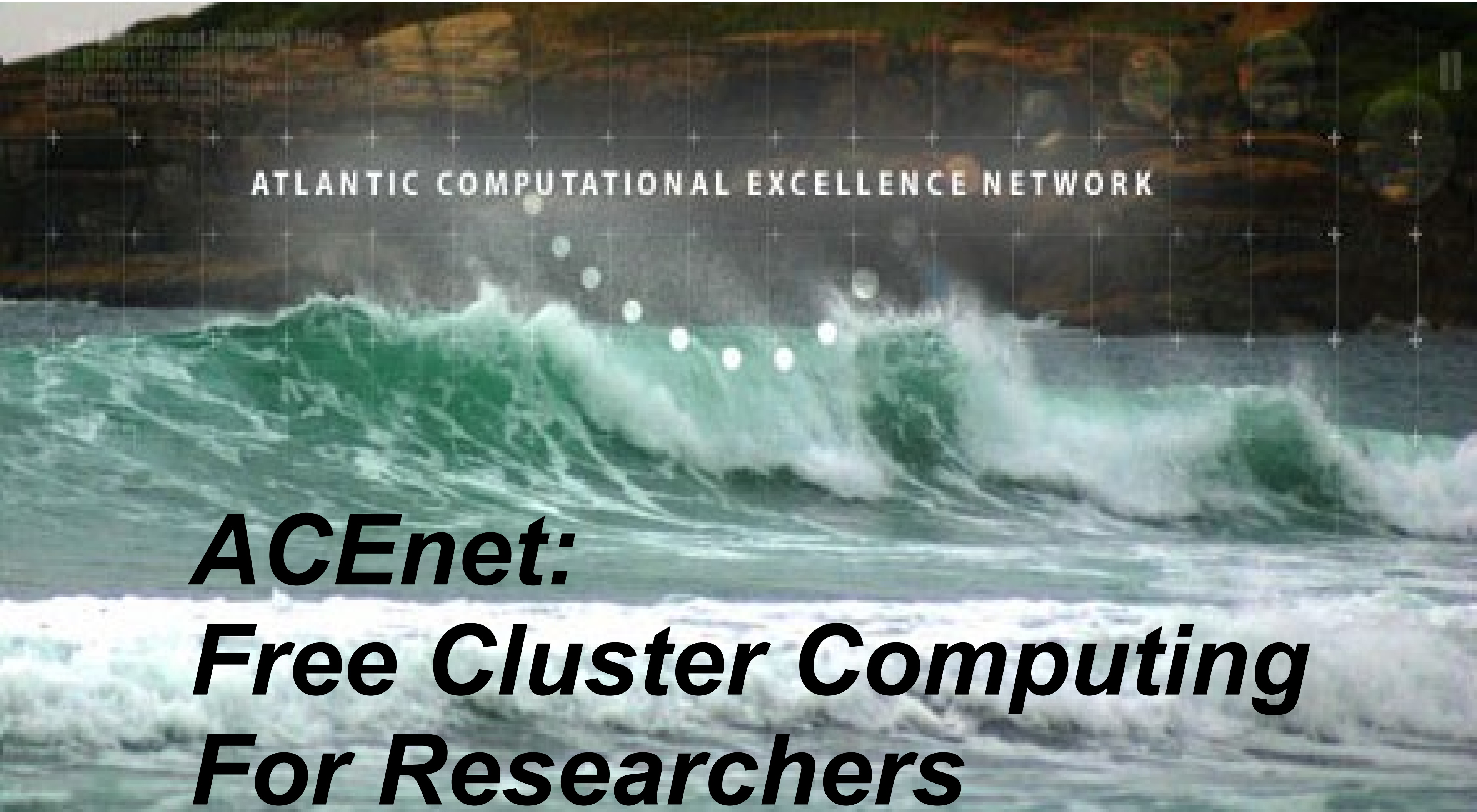


ACEnet: Free Cluster Computing For Researchers

Dr. Ross Dickson, Computational Research Consultant

What is ACEnet?

- <u>A</u>tlantic <u>C</u>omputational <u>E</u>xcellence <u>Net</u>work
- Shared regional resource for ...

  … high-performance computing (HPC)

  … remote collaboration

  … visualization
- Hardware, software, and personnel

Remote Collaboration

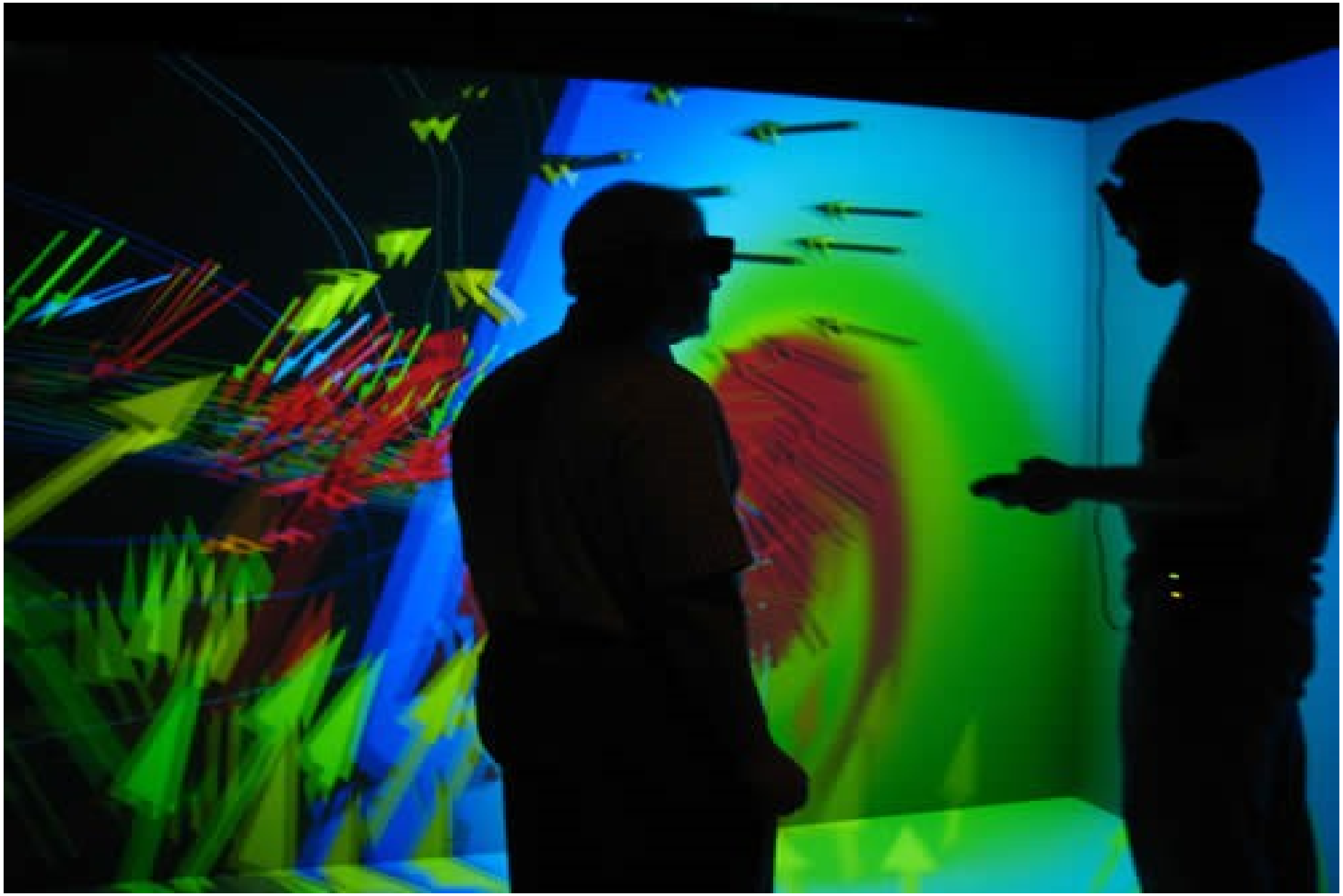
Internet videoconferencing - inSORS & Access Grid

Dalhousie - Memorial - Mount Allison - Saint Mary's - Saint Francis Xavier
Acadia - New Brunswick **+ other sites across Canada & the world**

Advanced Visualization

Data cave

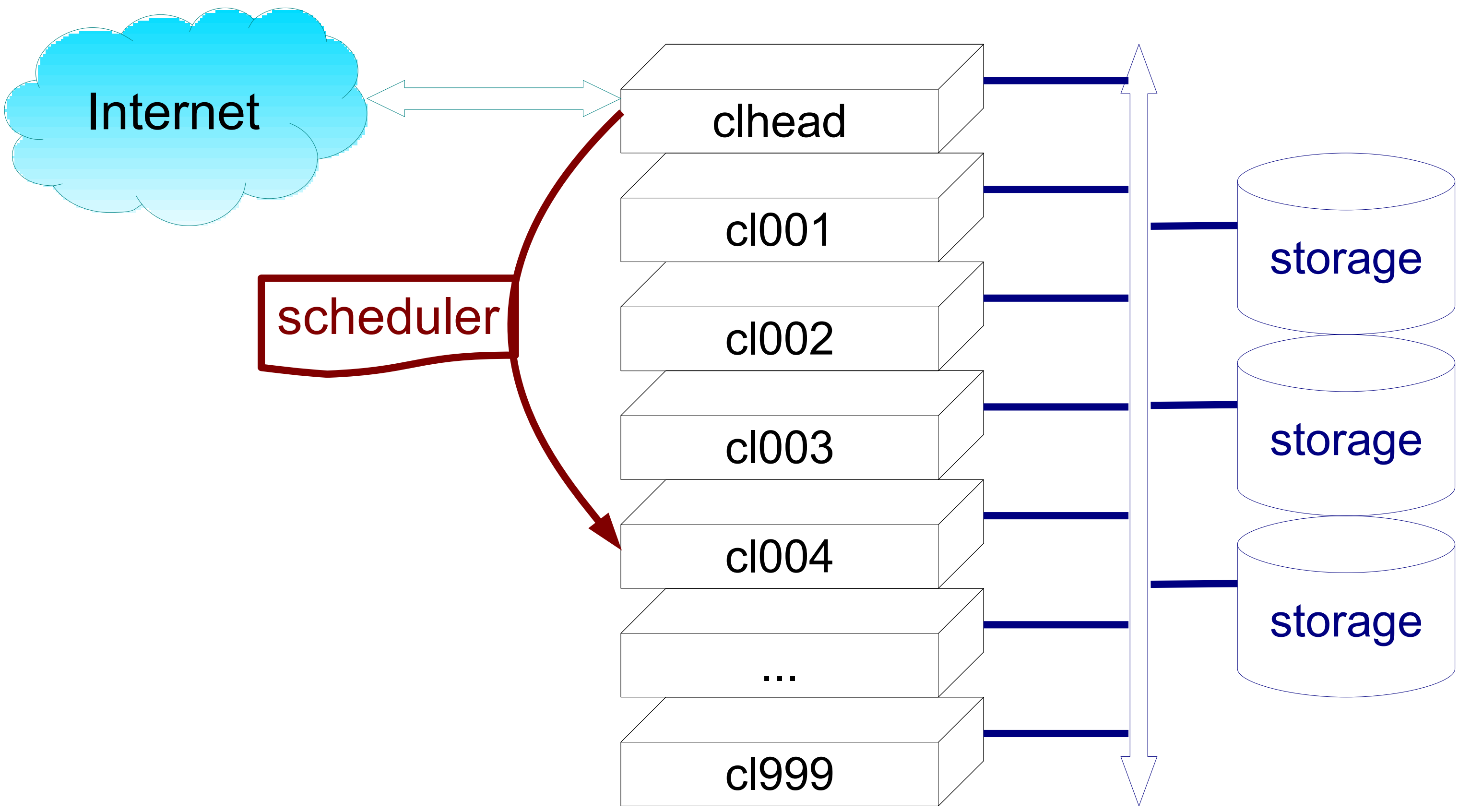
3D immersive data exploration facility **@ Saint Mary's**



What's “High Performance Computing”?

- Many CPUs working on one problem
  - Many “serial” jobs running at once, or
  - many CPUs working in close coordination.
    - Communication between CPUs is key!
- Advantages
  - Faster results
  - Larger problems or more detailed simulations
- Principal tool: Computer **cluster**
  - Racks of regular computers, networked

Cluster workings



CPU and Memory Configurations

ACEnet clusters — 6636 cores

Cluster	Nodes	Cores/node	RAM/node	Total cores
Brasdor	233	4	8G	932
Courtenay	13	4	16G	52
Fundy	38	16	64G, 128G	636
	7	4	16G	
Glooscap	55	4	4G	1580
	85	16	32G - 128G	
	40	owned by individual groups		640
Mahone	138	4	16G, 64G	552
Placentia	154	16	32G, 64G	2884
	17	12	24G	
	54	4	16G, 64G	
	117	owned by individual groups		908

Storage Configurations
and default quotas

Cluster	/home user quota	/globalscratch user quota	/nqs size	/scratch/tmp
Brasdor	47 Gb	238 Gb	14 Tb	20 - 330 Gb
Courtenay	--	--	2.0 Tb	100 Gb
Fundy	47 Gb	238 Gb	14 Tb	35 - 100 Gb
Glooscap	47 Gb		2.8 Tb	0 - 400 Gb
Mahone	13 Gb	238 Gb	14 Tb	50 - 100 Gb
Placentia	13 Gb	238 Gb	14 Tb	0 - 140 Gb
	Cluster-local			Node-local

Applications

http://www.ace-net.ca/wiki/Software

Molecular	**Math&CS**	**Engineering**	**Earth &Ocean**	**Bio**
Gaussian	Sage	ANSYS	CDO	AbySS
Gromacs	GAP	(Fluent)	NCO	Bowtie
CPMD	DiVinE	CFX	GMT	mpiBlast
Abinit	...	LS-DYNA	MM5	MrBayes
NAMD	**Astro**	elmer-fem	...	Migrate-n
WebMO	StarLab			PhyML
VMD	IRAF			PLINK
...	...			...

Software requests to:

support@ace-net.ca

Software Development

http://www.ace-net.ca/wiki/Software

Languages	**Compiler Suites**	**Parallel APIs**	**Libraries**	**Tools**
C/C++	GCC	Open MPI	LAPACK	TotalView
Fortran	Intel	OpenMP	FFTW	dbx, gdb
Perl	PGI	...	GSL	Valgrind
Python	SunStudio		ACML	make
R			netCDF	cvs, svn
Java			HDF	git
...			Boost	...
			...	

Software requests to:

support@ace-net.ca

Data Analysis

http://www.ace-net.ca/wiki/Software

R

Rmpi

Octave

Matlab*

*Distributed Computing Server

Python

Numpy

SciPy

Sage

pyMPI

pylab

matplotlib

Software requests to:

support@ace-net.ca

What's the cost?

- No money. It's paid for.
- Several $million from
  - Canada Foundation for Innovation (CFI)
  - Atlantic Canada Opportunities Agency (ACOA)
  - Provinces (NS, Nfld, NB)
  - “in kind” contributions from ~~Sun Microsystems~~ Oracle and hosting universities

No really: What's it cost?

- Keep publishing.
  - and tell us about it, once a year in conjunction with account renewal
- Help us out.
  - Local User Groups ← user direction!
  - Research Directorate, other committees
- Share the resource.
  - about 300 projects, 1000 users
  - jobs must use dynamic resource manager, "Sun Grid Engine"

Command-Line Interface

glooscap-head.cs.dal.ca - PuTTY

```
Using username "rdickson".
rdickson@glooscap.ace-net.ca's password:
Last login: Wed Mar 16 10:03:16 2011 from rdickson.ucis.dal.ca
Welcome to Glooscap.ace-net.ca - Halifax, Nova Scotia
****************************************************

rdickson@glooscap: ~ $ /usr/local/R/bin/R

R version 2.8.0 (2008-10-20)
Copyright (C) 2008 The R Foundation for Statistical Computing
ISBN 3-900051-07-0

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> 
```

Environment modules

```
$ module list
Currently Loaded Modulefiles:
 1) pgi/8.0-6   2) openmpi/pgi/1.2.9 ...
$ module avail
    ...long list ...
$ module load package
$ module help
```

Training Available

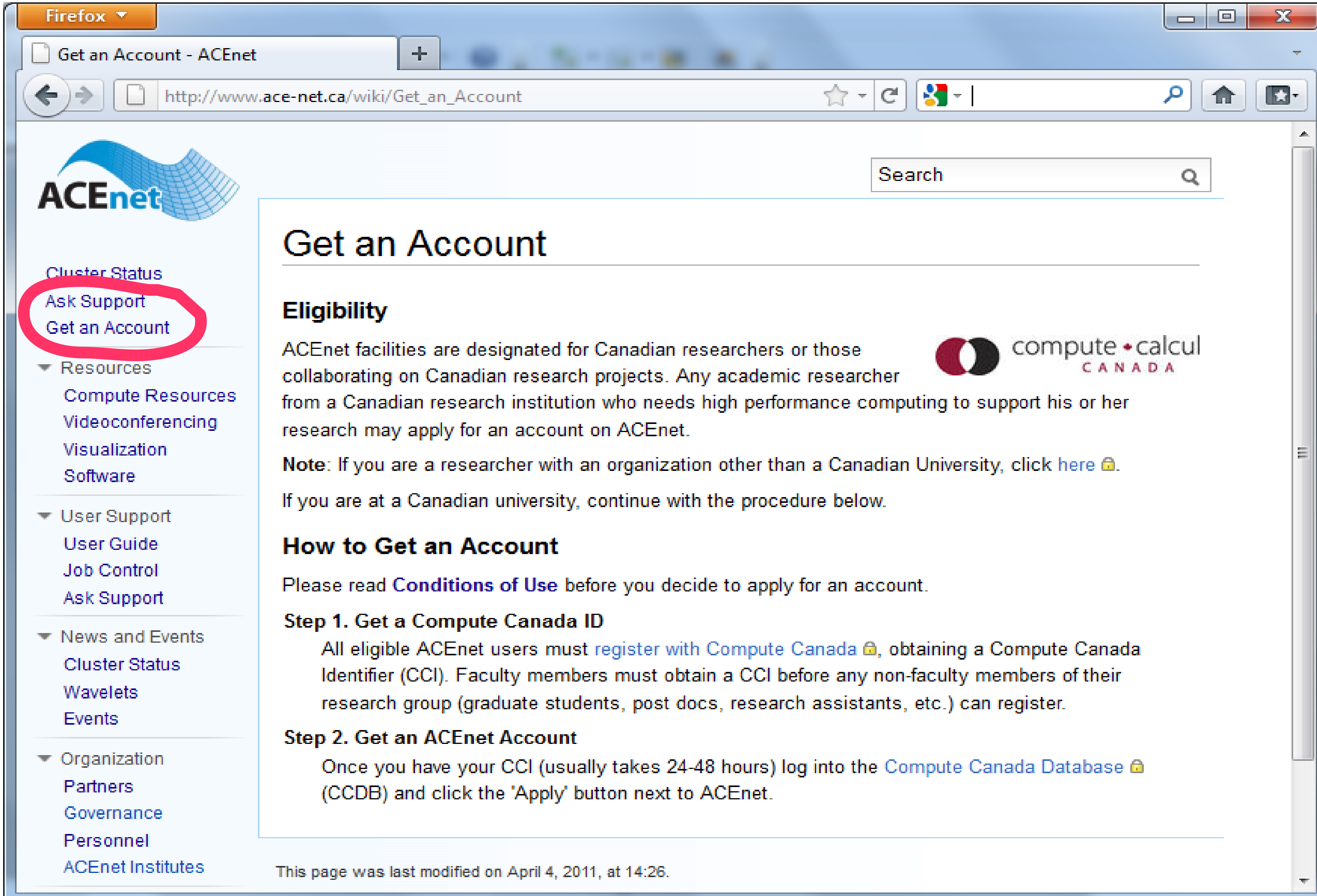
Seminars on demand (~ 1 hr each)

- Introduction to ACEnet
- Linux command line
- Using Sun Grid Engine
- Perl or Python Programming
- ...many more...

- Custom tutorials
  - "Managing Many Jobs" at request of a physics grp
  - ...*your interest here...*

- Scheduled seminar series
  - www.ace-net.ca/wiki/Events

Getting an account: *http://www.ace-net.ca/*



Get an Account

Eligibility

ACEnet facilities are designated for Canadian researchers or those collaborating on Canadian research projects. Any academic researcher from a Canadian research institution who needs high performance computing to support his or her research may apply for an account on ACEnet.

compute • calcul CANADA

Note: If you are a researcher with an organization other than a Canadian University, click here.

If you are at a Canadian university, continue with the procedure below.

How to Get an Account

Please read **Conditions of Use** before you decide to apply for an account.

Step 1. Get a Compute Canada ID

All eligible ACEnet users must register with Compute Canada, obtaining a Compute Canada Identifier (CCI). Faculty members must obtain a CCI before any non-faculty members of their research group (graduate students, post docs, research assistants, etc.) can register.

Step 2. Get an ACEnet Account

Once you have your CCI (usually takes 24-48 hours) log into the Compute Canada Database (CCDB) and click the 'Apply' button next to ACEnet.

This page was last modified on April 4, 2011, at 14:26.

How Do I get Connected?

- Logging in
  - ssh *Secure Shell*
- Moving Data Around
  - sftp *Secure File Transfer Protocol*
  - scp *Secure Copy*

- both provided with Mac OS X or Linux
  - ssh -X username@brasdor.ace-net.ca
  - sftp username@placentia.ace-net.ca

- Windows? Use PuTTY and WinSCP

  just google for them

Where to go for help

- ACEnet User Guide
  - http://www.ace-net.ca/wiki/User_Guide
- ACEnet User Wiki
  - http://www.ace-net.ca/wiki/ACEnet
- Research/user support personnel
  - email: support@ace-net.ca

What can we do for you?

What cluster should I use?

- Doesn't matter,
  ONE ACCOUNT IS GOOD EVERYWHERE

- Multiple serial jobs? Brasdor best
- Shared memory parallel? Fundy best
- MPI? Mahone best, Placentia, Glooscap, Fundy good
- Some software has only local license
  - *e.g.* Matlab DCS, Gaussian only at Placentia
  - See http://wiki.ace-net.ca for availability, or contact your Computational Research Consultant

Member institutions